Typology-Guided Image Captioning for Low-Resource Languages

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Abstract

Multilingual image captioning models achieve strong performance on high-resource languages but degrade significantly for low-resource languages (LRLs), largely due to limited training data. Linguistic typology offers a potential signal for bridging this gap, capturing structural similarities between languages that may enable cross-lingual transfer. In this work, it is investigated whether typological information can improve multilingual captioning by conditioning a vision-language model on URIEL typological vectors. A modular architecture is built combining a frozen BLIP-2 visual encoder with an mT5 decoder, and a strong multilingual baseline is established using an encoder-level language embedding for language steering. On top of this, typology is introduced through two mechanisms: Feature-wise Linear Modulation (FiLM) and typology-based prompt tokens. Across a controlled set of experiments on seven training languages and evaluation on XM3600, typological structure is found to be learnable: both FiLM parameters and prompt representations reflect meaningful cross-lingual similarities. However, this does not directly translate into improved generation. Unconstrained FiLM scaling destabilizes feature distributions and degrades performance, while zero-shot transfer to unseen languages fails under limited typological coverage. In contrast, prompt-based conditioning yields small but consistent improvements on seen languages without disrupting the underlying model, and the encoder-level language embedding remains the most effective and stable signal for multilingual generation. Overall, the results show that typology is a useful but delicate signal: its effectiveness depends critically on how it is integrated and on the diversity of training languages. It is concluded that broader typological coverage (i.e., more diverse languages rather than more data per language) and multilingual visual alignment are necessary for typology-driven transfer to be effective in low-resource multilingual captioning.

1. Introduction

Image captioning is the task of generating natural language descriptions of visual content. Recent multilingual vision-language models such as mBLIP (Geigle et al., 2023), UC2 (Zhou et al., 2021), and PaLI (Chen et al., 2023) can caption images across many languages from a shared backbone. However, performance remains highly uneven: high-resource languages achieve strong results, while low-resource languages (LRLs) degrade significantly or fail entirely (Geigle et al., 2023). This gap is largely driven by data scarcity, as captioning datasets are predominantly English or English-translated, with long-tail languages underrepresented (Thapliyal et al., 2022). In practice, LRL captioning relies either on training language-specific models, which do not scale (Choudhury et al., 2025; Kaur et al., 2021), or translating English captions, which often impose English structure and propagate errors. Linguistic typology offers a potential way to bridge this gap (Lin et al., 2019; Ponti et al., 2020; Maurya & Desarkar, 2023). Typology captures structural properties such as word order, case marking, and morphology. Since different languages require different structural realizations of the same visual scene, typological information could guide generation toward language-appropriate structure. For example, a model trained on English may produce captions that are lexically Bengali but structurally English; typology could help correct this mismatch. This idea is investigated by building a multilingual captioning system that conditions generation on typological features. The architecture uses a frozen BLIP-2 (Li et al., 2023) visual encoder, with Q-Former features mapped into an mT5-base (Xue et al., 2021) decoder via a learned projection layer. This forms a typology-agnostic baseline, augmented with an encoder-level language embedding for language identity. Typology is then introduced through two mechanisms: Feature-wise Linear Modulation (FiLM) (Perez et al., 2018) using URIEL vectors (Littell et al., 2017), and typology-based prompt tokens. This setup allows us to pose two research questions:

RQ1: Does typological conditioning improve caption quality on seen languages compared to a typology-agnostic baseline?

RQ2: Does typological proximity enable zero-shot transfer to unseen languages, and what are its limitations?

Our contributions are threefold. (1) A typology-conditioned multilingual captioning framework is proposed with FiLM and prompt-based integration, evaluated against a strong

baseline on XM3600. (2) It is shown that typological structure is learnable, but unconstrained FiLM destabilizes feature distributions, while prompt-based conditioning provides modest but consistent gains. (3) It is found that encoder-level language embeddings remain the most effective signal, and that zero-shot transfer fails under limited typological coverage.

2. Related Work

Multilingual Image Captioning. Several approaches have addressed image captioning beyond English. The most direct is to build a separate captioning model for each target language, as in Punjabi (Kaur et al., 2021) and Assamese (Choudhury et al., 2025). While effective for individual languages, this approach requires a dedicated dataset and model for each new language and does not scale. A common alternative is to generate English captions and translate them into the target language (Wu et al., 2019), but this often preserves English information structure, which is problematic for languages with different word order or case marking systems. More recent work has focused on training a single multilingual model. mBLIP (Geigle et al., 2023) attaches a multilingual text decoder (mT0-XL) to a frozen BLIP-2 vision encoder and re-aligns the Q-Former using 2.5 million multilingual image-text pairs obtained through machine translation. This produces a system that can caption in 95 languages from a shared backbone. Similarly, UC2 (Zhou et al., 2021) and PaLI (Chen et al., 2023) achieve broad language coverage through large-scale multilingual pretraining, but at substantial computational cost. A complementary direction is retrieval-augmented captioning: LMCAP (Ramos et al., 2023) and ConCap (Sherif et al., 2025) retrieve captions of visually similar images from a multilingual corpus and use them as in-context examples, achieving competitive XM3600 results with little or no training data. All of these approaches rely on training or retrieval data to capture language variation rather than explicitly conditioning generation on the structural properties of the target language. As a result, performance on low-resource languages remains tied to data availability, with no mechanism to exploit structural similarities between related languages. To the best of our knowledge, typological conditioning has not been explored at the vision-language interface, where the challenge is not only linguistic variation but also how visual semantics are structured differently across languages.

Typology-Aware NLP. Linguistic typology has been used in several NLP settings to improve cross-lingual transfer. MetaXNLG (Maurya & Desarkar, 2022), for example, uses URIEL features (Littell et al., 2017) to cluster languages and schedule meta-learning episodes for text-only generation, but typology is not fed directly to the model as an input signal. The present work differs in that typology is used as a continuous conditioning signal at generation time: the URIEL vector for each language is transformed into FiLM parameters that directly modulate the model's internal representations, shaping how visual information is structured before decoding. This shifts typology from a training-time organisational signal to a runtime mechanism that influences generation behaviour. Closer to this approach, Üstün et al. (2022) use language-level typological features to generate adapters for neural machine translation, allowing a single model to handle multiple language pairs without separate fine-tuning. Their work operates in a text-to-text setting, whereas a similar principle is applied at the vision-language interface by conditioning visual features before they enter a multilingual decoder. Ansell et al. (2021) similarly generate task-specific adapters from language embeddings for cross-lingual transfer. These works demonstrate that typological information can serve as a useful inductive bias for cross-lingual generalization, and this idea is extended to a multimodal captioning setting.

Data Scarcity in Multilingual Captioning. Data scarcity is more severe for image captioning than for text-only tasks, because it requires paired image-text data rather than monolingual text alone. For many low-resource languages, no captioning datasets exist at all, and the few that do tend to be small and domain-specific (Thapliyal et al., 2022). Methods such as upsampling, data augmentation, and balanced training schedules (Wang et al., 2020; McGiff & Nikolov, 2025) can help to some extent, but they cannot create structural knowledge about a language that the model has never seen. This motivates approaches like ours, which aim to transfer structural knowledge across languages using typological information, enabling the model to generalise based on linguistic structure rather than solely on data availability.

3. Dataset and Task

3.1. Task Definition

The task is multilingual image captioning: given an input image x and a target language L, the model generates a caption y_L describing the image in language L. This project primarily focuses on low-resource languages (LRLs) with several high-resource languages as controls. Two settings are evaluated: *seen* languages, where the model has access to training captions in that language, and *zero-shot* languages, where the model must generate captions in a language never encountered during training. CIDEr (Vedantam et al., 2015) is used as the primary metric, which measures consensus-based n-gram overlap with human references weighted by TF-IDF, rewarding specific and reference-like captions over generic ones. Models are also evaluated using BLEU-1 and BLEU-4 (Papineni et al., 2002) for comparability with prior work.

3.2. Training Datasets

A multilingual training corpus is assembled spanning seven languages chosen to cover a range of typological families and resource levels. The datasets vary in size, annotation method, and image source, reflecting the practical reality of working with multilingual captioning data. Table 1 summarizes the key statistics.

Lang	Dataset	Size	Source	Type
English	Multi30K	31K	Flickr30k	Crowdsourced
German	Multi30K	31K	Flickr30k	Prof. MT
Ukrainian	Multi30K-uk	29K	Flickr30k	MT + human
Arabic	Flickr8k-ar	24K	Flickr8k	MT + human
Turkish	TasvirEt	16K	Flickr8k	Human
Bengali	BAN-Cap	40K	Flickr8k	Human
Vietnamese	UIT-OpenViIC	61K	Own imgs	Human
Total		**233K**		

Table 1. Training datasets. Both the typology-agnostic baseline and the typology-conditioned model are trained on the same data.

Dataset descriptions. Both human-written and translated captions are combined. English and German are built via Multi30K (Elliott et al., 2016), where English captions are crowdsourced and German captions are professionally translated. Ukrainian caption sets are built using Multi30K-uk (Saichyshyna et al., 2023), generated via machine translation followed by human correction. For Arabic, the Arabic Flickr8k is used (ElJundi. et al., 2020), which follows a similar pipeline with professional verification. For Turkish (TasvirEt (Unal et al., 2016)), Bengali (BAN-Cap (Khan et al., 2022)), and Vietnamese (UIT-OpenViIC (Bui et al., 2023)), these datasets are fully human-annotated datasets, with Vietnamese additionally using images collected from native-language queries. Overall, the corpus contains typologically varied language resources with diverse annotation methods.

Preprocessing. The data is split into training and validation sets using a 90/10 image-level split with a fixed random seed (42).

The datasets are substantially imbalanced, ranging from 16,037 captions for Turkish to 61,241 for Vietnamese. Without any correction, this imbalance causes the model to be dominated by the larger languages, to the point where it generates Vietnamese captions regardless of the target language. To address this, a `WeightedRandomSampler` is used that assigns each sample a weight inversely proportional to the size of its language. This produces approximately 14,500 samples per language per epoch (determined by the smallest dataset, Turkish), for a total of roughly 101,000 samples per epoch.

3.3. Evaluation: XM3600

For evaluation, the Crossmodal-3600 (XM3600) benchmark is used (Thapliyal et al., 2022), which provides 3,600 geographically diverse images with human-written captions in 36 languages. Each image has one or more reference captions written by native speakers rather than translated from English. No XM3600 data is used during training.

XM3600 images depict culturally diverse scenes (local landmarks, food, festivals) that differ substantially from the Flickr-based training data. This domain gap is expected to lower absolute scores, though relative comparisons between models remain valid.

3.4. Language selection via URIEL clustering

URIEL and typological feature vectors.

The URIEL typological database (Littell et al., 2017) provides language representations derived from the World Atlas of Language Structures (WALS, Dryer & Haspelmath 2013).

Missing values are imputed using the `syntax_knn` fallback, which estimates features from typologically similar languages. Across the 7 training languages, Bengali has the highest imputation rate at 51 of 103 dimensions, reflecting limited WALS documentation of Indic languages (also discussed in Section 5). Arabic has the lowest imputation rate at 5 of 103 dimensions.

All vectors are z-score normalised, this places all languages in a shared continuous typological space, enabling interpolation between seen and unseen languages.

Clustering methodology. To identify typologically coherent language groups among the 36 XM3600 languages, K-means clustering is applied (k=8) to L2-normalised `syntax_average` vectors using cosine distance. Stability is assessed over 15 random initialisations: a pair's stability score is the fraction of runs in which they co-occur in the same cluster. Pairs with stability below 0.5 are considered unreliable.

Clustering results. Table 2 summarises the resulting clusters.

Cluster	Members	Dist	Stability
Germanic	en, de, nl, sv, da, no	0.082	1.00
Slavic	uk, ru, pl, cs	0.111	0.87–1.00
Romance+Semitic	ar, he, fr, es, it, pt, ro, el	0.142	0.73–1.00
SE Asian	vi, id, th, mi	0.217	0.87–1.00
Indo-Aryan/Dravidian	bn, hi, te	0.191	1.00
Turkic/Uralic	tr, hu, fi	0.209	0.40–0.93
Sino-Tibetan	zh, ja, ko	0.236	1.00
Other	fa, sw	0.378	0.33

Table 2. URIEL clusters on XM3600 languages.

Cluster compactness varies significantly: Germanic is tightly grouped (0.082, with Dutch–German at 0.030), Slavic is moderate (0.111, Ukrainian–Czech at 0.134), while Indo-Aryan/Dravidian is substantially more dispersed (0.191). Bengali–Telugu at 0.177 and Bengali–Hindi at 0.178. Despite belonging to the same cluster, Indo-Aryan/Dravidian languages are more than twice as distant as Germanic pairs. This asymmetry directly affects zero-shot transfer under FiLM conditioning. (Experiments 5.3). The FiLMGenerator is trained on 7 languages and must generalise to zero-shot languages by extrapolating from the nearest training language in URIEL space. Dutch and Czech are genuinely close to their respective training languages (German and Ukrainian), so the FiLMGenerator can interpolate rather than extrapolate. In contrast, Hindi and Telugu, despite clustering with Bengali, lie outside the convex hull of the 7 training languages in 103-dimensional URIEL space. Cluster membership motivates the experi-

mental design, but intra-cluster distances provide a more precise predictor of transfer performance.

Training language selection. From each stable cluster, one training language is selected based on data availability: English and German, Ukrainian, Arabic, Vietnamese, Bengali, and Turkish. This gives 7 training languages covering 5 of the 8 clusters, with 10 zero-shot evaluation languages drawn from within or adjacent to those clusters.

4. Methodology

4.1. Base Architecture and English Pretraining

Visual encoding.

BLIP-2 (Li et al., 2023) is used as a frozen visual encoder, producing a fixed sequence of 32 query embeddings of dimension 768. Visual features are precomputed offline for all images.

Projection layer. A two-layer MLP connects the Q-Former output to the mT5 encoder input space. The ProjectionMLP applies a linear transformation (768→768), GELU activation, layer normalisation, and a second linear transformation (768→768), totalling 1,182,720 parameters. It is trained alongside the LoRA adapters from the English pretraining stage.

Text generation. mT5-base (Xue et al., 2021) is used as the multilingual text decoder, adapted with Low-Rank Adaptation (LoRA, Hu et al. 2022) .

English pretraining. The ProjectionMLP and LoRA adapters are first trained on English Flickr8k to establish visual-language alignment, producing a base English checkpoint from which all subsequent multilingual models are initialised.

4.2. Multilingual Baselines

O1-lang: encoder-level language identity. To extend the model to seven languages, a language embedding layer is introduced: a learned `nn.Embedding(7, 768)` that maps a discrete integer language identifier to a 768-dimensional dense vector. At each forward pass, this vector is prepended to the projected visual features before they are passed to the mT5 encoder, producing an encoder input of shape $(B, 33, 768)$:

$$\text{enc} = [\, \mathbf{e}_{\text{lang}} \,\big|\, \text{proj}(F) \,] \in \mathbb{R}^{B\times 33\times 768}, \tag{1}$$

where $\mathbf{e}_{\text{lang}} \in \mathbb{R}^{768}$ is the language embedding and $\text{proj}(F)$ are the projected visual tokens. The decoder cross-attends to all 33 positions at every generation step, meaning the language identity signal is continuously available throughout generation rather than only at the first token. Training captions are prefixed with a short native-script string per language (e.g., `Deutsch:` for German). At inference, the first decoded token is constrained to match the first subword unit of this prefix using the `forced_bos_token_id` parameter of the mT5 generation API, which forces a specific token at decoding step 1 without restricting subsequent steps.

The ProjectionMLP, LoRA adapters, and language embedding are trained jointly from the English checkpoint using AdamW (lr = 1×10^{-4}, weight decay 10^{-2}, cosine annealing over 10 epochs, batch size 16).

O1-bos: decoder-level language steering. As an ablation of O1-lang, a variant is trained in which the language embedding is removed entirely and only the label prefix at training time and the `forced_bos` mechanism at inference is used. The encoder input is thus $(B, 32, 768)$, 32 visual tokens only, with no explicit language token.

4.3. FiLM Typological Conditioning

Motivation. It is hypothesised that conditioning the visual features on typological properties of the target language: word order, case marking, morphological complexity, may allow the model to emphasise different aspects of the visual scene depending on how the target language structures descriptions.

FiLM conditioning. Feature-wise Linear Modulation (FiLM, Perez et al. 2018) was introduced for visual reasoning: in the original VQA setting, a question vector generates per-channel scale and shift parameters that modulate intermediate ResNet feature maps, allowing the network to attend differently to visual features depending on the question being asked. The same mechanism is applied at the visual-language interface. FiLM allows the model to "reweight" visual features depending on the target language. For example, it can amplify features relevant to word order or suppress features that are less important for a given language. The URIEL typological vector generates per-dimension scale (γ) and shift (β) parameters applied to the projected visual features before they reach the mT5 encoder:

$$\tilde{F} = \gamma \odot F + \beta, \tag{2}$$

where $F \in \mathbb{R}^{B\times 32\times 768}$ are the projected visual tokens and $\gamma, \beta \in \mathbb{R}^{B\times 768}$ are broadcast across the 32 sequence positions. Thus, FiLM modifies the scale and offset of the same visual token sequence rather than introducing new tokens.

The FiLMGenerator maps the 103-dimensional URIEL vector to (γ, β):

$$\mathbb{R}^{103} \xrightarrow{\text{Linear}} \mathbb{R}^{128} \xrightarrow{\text{ReLU}} \xrightarrow{\text{Linear}} \mathbb{R}^{1536}, \tag{3}$$

with the output split into γ and β of 768 dimensions each (211,456 parameters total). The final linear layer is initialised to output γ=**1** and β=**0** everywhere, so at epoch 0 the FiLM operation is an exact identity and the model is equivalent to O1-lang (Figure 1).

4.4. Typology-Based Prompting

Motivation. Prompt-based methods have been shown to effectively influence pretrained models without modifying their parameters (Lester et al., 2021). It has also been shown to be efficient for image captioning, handling complex images (Cui & Cui, 2024). As demonstrated by Yang et al. (2025) (Yang et al., 2025), language-specific information

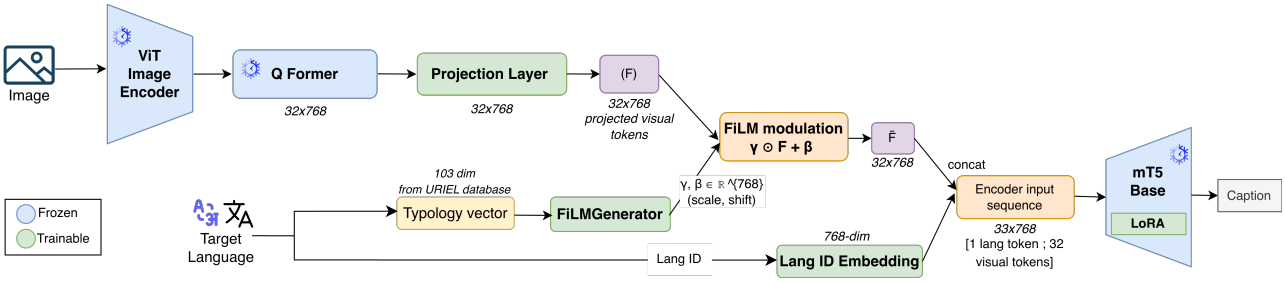


Figure 1. FiLM-based typological conditioning. A URIEL vector generates per-dimension scaling (γ) and shifting (β) parameters that modulate projected visual features before encoding.

can be encoded in prompts. This idea is extended to multilingual captioning by encoding typological information as input prompts. FiLM is replaced with a typology-based prompt generator that maps language features into learnable tokens prepended to the encoder sequence. The decoder can selectively attend to without modifying any existing model weights. This enables typological information to guide generation through attention rather than feature transformation. As seen in Figure 2, a `TypologyPromptGenerator` maps the 103-dimensional URIEL vector into k=8 dense prompt tokens via a two-layer MLP with GELU activation and layer normalisation (103→256→8×768). These tokens are prepended to the encoder input sequence, forming:

$$\text{enc} = [\, p_1 \cdots p_8 \,|\, \mathbf{e}_{\text{lang}} \,|\, \text{proj}(F) \,] \in \mathbb{R}^{B\times 41\times 768}, \quad (4)$$

where the input sequence consists of prompt tokens ($p_1 \cdots p_8$), a language embedding ($\mathbf{e}_{\text{lang}}$), and projected visual features from the visual encoder (proj(F)). These prompt tokens are used as additional key-value memory in the decoder's cross-attention. This allows typological information to influence generation through attention, without altering the underlying visual feature distribution.

Training. Unlike O2, the LoRA adapters are unfrozen and trained jointly with the TypologyPromptGenerator. This enables the decoder to learn how to incorporate the injected prompt tokens during training. As a result, the typological information they encode can influence the generated output.

5. Experiments

Four experiments are presented to address the research questions. Experiment 5.1 isolates the contribution of the language embedding to multilingual generation quality. Experiment 5.2 evaluates whether FiLM conditioning on URIEL vectors improves performance on seen and zero-shot languages (RQ1, RQ2). Experiment 5.3 analyses the internal behaviour of the trained FiLMGenerator to explain the observed results. Experiment 5.4 tests whether a prompt-based mechanism can overcome the limitations with FiLM.

5.1. Language Steering Ablation (O1-lang vs O1-bos)

Encoder-level language embeddings (O1-lang) are compared with decoder-level forced tokens (O1-bos). Both models share the same architecture, training data, and hyperparameters.

Results and Discussion. O1-lang outperforms O1-bos on every language by a factor of approximately 1.7–2.5× in CIDEr (Table 3). The gains are consistent across all languages and particularly pronounced for Vietnamese (+6.09) and English (+5.78), indicating that encoder-level conditioning provides a strong and reliable signal for multilingual generation. This demonstrates that providing a persistent language signal at the encoder level is critical for multilingual caption generation, as opposed to relying on a single decoder-level constraint.

	CIDEr		Len ratio	
Lang	**bos**	**lang**	**bos**	**lang**
en	9.71	**15.49**	1.291	1.121
de	3.01	**5.76**	1.109	1.025
uk	0.91	**2.14**	1.207	1.067
ar	2.13	**5.40**	1.254	1.030
tr	1.57	**3.63**	1.244	1.143
bn	2.49	**4.47**	1.041	0.998
vi	4.37	**10.46**	0.822	0.898

Table 3. Language steering ablation on XM3600. O1-lang vs O1-bos.

O1-bos over-generates substantially more than O1-lang across all languages. The decoder in O1-bos receives no language signal after the first token and defaults to the dominant training language pattern. In contrast, O1-lang maintains better length calibration, suggesting that continuous access to language identity helps regulate both content and structure during generation. O1-lang is the primary multilingual baseline for all subsequent experiments; the lang_embedding is a useful architectural contribution.

5.2. FiLM Conditioning Effect (O1 vs O2)

It is tested whether FiLM conditioning provides a stronger conditioning signal than the encoder-level language embedding established in Experiment 1.

Zero-shot inference. For languages not seen during training, inference uses two variants. Variant A applies no FiLM and uses forced_bos only for language steering. Variant B applies FiLM conditioned on the zero-shot language's URIEL vector alongside forced_bos, with a zero vector replacing the lang_embedding (which was only trained for the seven seen languages).

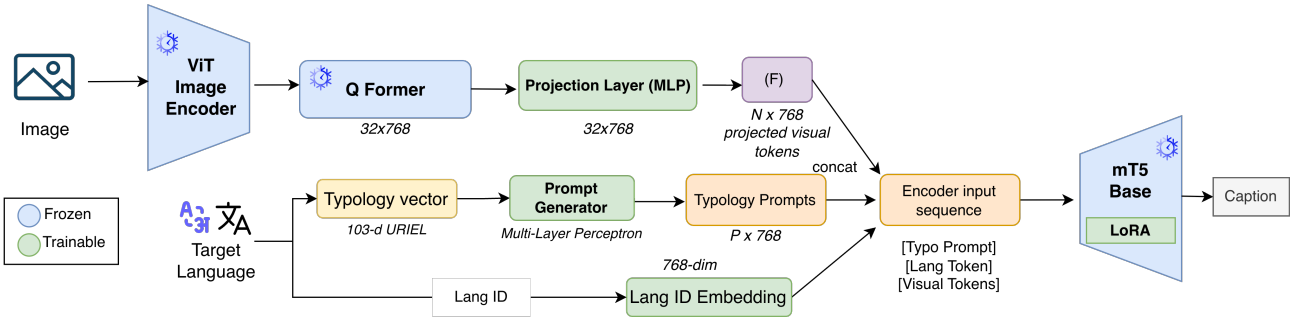


Figure 2. Typology-based prompting. A URIEL vector is mapped to prompt tokens that are prepended to the encoder input sequence for conditioning.

Results. Table 4 shows CIDEr and BLEU scores for all 7 seen languages. Zero-shot CIDEr is identical between O1 and O2: nl=0.11, cs=0.01, id=0.02, mi=0.04, he/hi/te/pl/hu/sw≈0.00. These near-zero scores indicate that neither model produces meaningful captions in unseen languages under this setup.

	CIDEr					BLEU-4		
Lang	**O1**	**O2**	Δ_F	**O3**	Δ_P	**O1**	**O2**	**O3**
en	15.49	14.89	−0.60	16.38	+0.89	4.45	4.25	4.59
de	5.76	5.02	−0.74	6.10	+0.34	1.33	1.13	1.50
uk	2.14	2.04	−0.10	2.29	+0.15	0.24	0.19	0.00
ar	5.40	4.97	−0.43	5.96	+0.56	0.51	0.47	0.56
tr	3.63	3.05	−0.58	3.67	+0.04	0.32	0.34	0.46
bn	4.47	4.18	−0.29	5.00	+0.53	0.34	0.00	0.30
vi	10.46	10.11	−0.35	10.79	+0.33	5.21	4.83	5.41

Table 4. Seen-language performance on XM3600 (3600 images). O1 is the encoder-language baseline, O2 adds FiLM conditioning, and O3 uses typology-based prompting. Δ_{FiLM} = O2 − O1 and Δ_{Prompt} = O3 − O1 are CIDEr differences relative to O1.

Discussion. FiLM conditioning under unconstrained γ (the per-dimension scaling parameter) degrades performance on every seen language. FiLM rescales each feature dimension of the projected visual representation. When γ grows too large, it amplifies features beyond the range the frozen decoder was trained to handle. The degradation is observed to be systematic rather than noisy, and correlates with the magnitude of γ, which grows from ~1.5 at epoch 1 to ~4–5× by epoch 10. This corresponds to a substantial shift in the feature distribution seen by the decoder. The model effectively "over-amplifies" visual features, pushing inputs into an unfamiliar regime and leading to degraded outputs. Bengali exhibits the most severe degradation, with BLEU-4 collapsing to zero and a 23.7% drop in unigram precision, consistent with the largest observed γ growth and sparse URIEL coverage (51/103 dimensions imputed).

5.3. Experiment 3: FiLM Behaviour Analysis

Motivation. The following is investigated: (a) did FiLM learn a typologically meaningful mapping; (b) why γ growth causes degradation and what is the principled bound. In particular, it is analysed whether the learned FiLM parameters reflect cross-lingual structure, and whether their magnitude explains the performance degradation observed in Experiment 2.

3a. γ/β Parameter Analysis

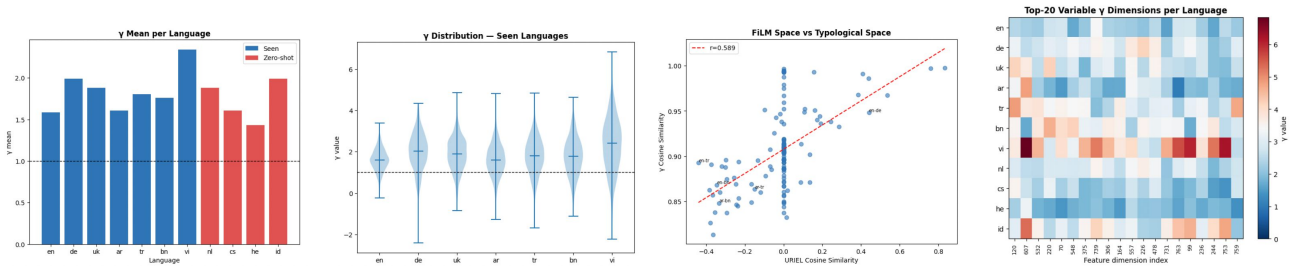
γ and β vectors (768-dim each) are extracted for all 17 languages (7 seen in training, 10 zero-shot) by passing their URIEL vectors through the trained FiLMGenerator.

Despite training on only 7 languages, FiLM learns a typologically coherent mapping. Pairwise cosine similarities between γ vectors correlate with URIEL typological similarities at r=0.589 (Figure 3c). Family-level clusters are correctly recovered without explicit supervision: German–Dutch (γ_{sim}=0.998, URIEL$_{\text{sim}}$=0.833) . The β values remain near zero for all languages throughout training (means between −0.075 and +0.007): the FiLM effect is entirely multiplicative, and a γ-only FiLM would halve output parameters at no performance cost.

Six zero-shot languages (Polish, Hungarian, Telugu, Hindi, Maori, Swahili) produce catastrophically large γ values (1.8–4.8×10^6). These extreme values correspond to the outliers visible in Figure 3a and Figure 3d. Their URIEL vectors fall outside the convex hull of the 7 training languages in 103-dimensional space, activating weight combinations in the second linear layer that were never constrained. The model is being asked to generalize to languages that are structurally unlike anything it has seen during training, forcing it to extrapolate rather than interpolate. Non-exploded zero-shot languages (Dutch, Czech, Hebrew, Indonesian) all fall near a training language: Dutch near German (URIEL$_{\text{sim}}$=0.833), Czech near Ukrainian (0.440). This links directly to the clustering analysis (Table 2).

3b. Feature Distribution and γ Constraint

The O1 projected feature distribution is measured on XM3600: mean=0.004, std=1.551, range [−5.53, 5.55]. This represents the distribution that the frozen decoder was trained to process. With γ mean reaching 4–5 by epoch 10, modulated feature std grows to ~6–8: approximately 4× beyond the frozen backbone's training distribution. In effect, FiLM amplifies features far outside the regime the model was trained on. The frozen LoRA adapters and



(a) γ mean per language (blue = seen, red = zero-shot).

(b) γ distribution for seen languages.

(c) Pairwise γ similarity vs URIEL similarity (r=0.589).

(d) Top-20 variable γ dimensions.

Figure 3. FiLM parameter analysis (non-exploded languages).

lang_embedding have no mechanism to compensate for this shift. This explains the systematic degradation observed in Experiment 2: the decoder receives inputs that are statistically out-of-distribution. A principled upper bound follows:

$$\gamma_{\max} = \frac{3\,\sigma_{\text{feat}}}{\sigma_{\text{feat}}} = 3.0, \qquad \beta_{\max} = \sigma_{\text{feat}} = 1.55, \qquad (5)$$

ensuring modulated features remain within the original 3σ range. Clamping $\gamma \in [0, 3.0]$ also prevents negative γ values (present in the unconstrained run), which invert individual feature dimensions, an operation the frozen model was never trained to handle. This constraint is derivable for any frozen pretrained model and is independent of the specific architecture or conditioning signal.

5.4. Experiment 5: Typology-Based Prompting (O3)

Figure 4 provides a four-part analysis of the learned prompt representations. Figure 4a shows that seen languages (e.g., en, de) exhibit larger deviations from zero, whereas zero-shot languages remain closer to the origin. This indicates that prompts encode stronger conditioning signals for seen languages, while remaining near-neutral for unseen ones. The prompt norm distribution (Figure 4b) reinforces this pattern: seen languages exhibit substantially higher norms (40–60) than zero-shot languages (15–20), indicating stronger activation of the prompt signal under supervision. Figure 4c shows a moderate correlation between prompt similarity and URIEL similarity ($r = 0.694$), indicating that typological structure is partially captured. However, the scatter remains diffuse, with many pairs deviating from the diagonal, suggesting inconsistent alignment across languages. The high-variance prompt dimensions (Figure 4d) show strong, structured activations for seen languages, but weaker and more uniform responses for zero-shot languages, indicating that typological information is encoded but not consistently activated. Overall, the model learns typology-aligned representations, but this structure is incomplete and fails to generalise effectively to zero-shot languages. To further examine how effectively typological information is learned and implemented, pairwise comparisons of prompt and typological similarity are presented in Table 5.

Language Pair	Prompt Sim	URIEL Sim
en–de	0.871	0.85
uk–cs	0.861	0.82
vi–id	0.969	0.90
ar–he	0.845	0.88
en–nl	0.910	0.92
ar–vi	0.628	0.02
tr–vi	0.322	-0.35
de–ar	0.178	-0.38
vi–cs	-0.140	0.00

Table 5. Prompt and typological similarity.

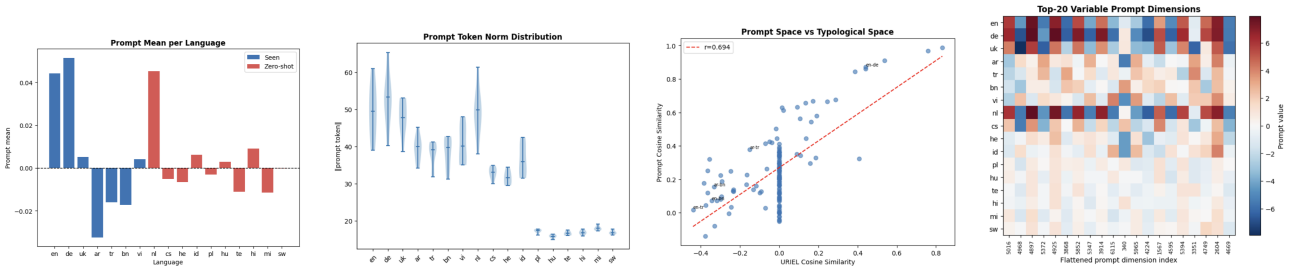
Typologically similar pairs (e.g., en–de, uk–cs, vi–id) exhibit high similarity in both spaces, confirming that the learned prompts encode meaningful linguistic structure. However, this similarity does not hold for all pairs. Some pairs (e.g., ar–vi, tr–vi, de–ar) exhibit high prompt similarity despite low typological similarity. In vi–cs case, the prompt similarity is negative while the typological similarity is zero. This means the model may only partially learn typologies of some languages, which may limit its ability to utilise such information effectively for caption generation.

Table 4 shows that typology-based prompting (O3) yields consistent but modest improvements across most seen languages. These gains are small (generally < +1 CIDEr) and uneven across metrics. BLEU-4 improvements are inconsistent and occasionally degrade, indicating limited impact on fine-grained generation quality. Taken together, these results show that while the model learns typological structure, this does not translate into substantial improvements in caption quality. Typology provides a meaningful inductive signal, but is insufficient on its own to drive strong multilingual generalisation.

6. Results and Evaluation

Table 6 summarises performance across model variants and provides context relative to mBLIP (Geigle et al., 2023). The best results (O3: CIDEr 16.38 for English, 6.10 for German) remain substantially below those of mBLIP (77.53, 30.66), reflecting architectural differences: mBLIP retrains the Q-Former on large-scale multilingual data, whereas the model uses a frozen, English-trained visual encoder.

RQ1: Effect of typological conditioning on seen lan-



(a) Mean prompt value per language (blue: seen, red: zero-shot).

(b) Prompt token norm distribution across languages.

(c) Prompt similarity vs. URIEL similarity (Pearson $r = 0.694$).

(d) Top-20 prompt dimensions by cross-language variance.

Figure 4. Prompt-based typological analysis across languages.

guages. Encoder-level language identity (O1-lang) significantly improves performance over decoder-only control (O1-bos), confirming that persistent language conditioning is necessary for multilingual generation. Soft prompt conditioning (O3) further yields consistent but modest improvements over O1-lang across most languages (e.g., +0.89 CIDEr for English, +0.56 for Arabic, +0.53 for Bengali). These gains indicate that typological signals are informative, but their impact remains limited within the constraints of a frozen backbone. In contrast, FiLM (O2) degrades performance across all languages, despite learning meaningful typological structure (Section 5.3), due to instability introduced by unconstrained feature scaling.

RQ2: Zero-shot transfer. Zero-shot performance remains near zero across all models and languages. Even for typologically close languages (e.g., Dutch, Indonesian), outputs collapse to dominant training-language patterns, with marginal scores arising from accidental n-gram overlap rather than genuine target-language generation. This indicates that typological conditioning alone is insufficient for zero-shot transfer in the absence of target-language exposure.

Summary. Typological information is learnable and partially useful, but its effectiveness depends on how it is integrated. Encoder-level conditioning provides a strong baseline, FiLM introduces instability under unconstrained scaling, and prompt-based conditioning offers a more stable but limited improvement. Effective transfer requires broader typological coverage during training or a multilingual visual encoder that produces language-agnostic features.

Model	en	de	ar	vi
O1-bos (ours)	9.71	3.01	2.13	4.37
O1-lang (ours)	15.49	5.76	5.40	10.46
O2 FiLM (ours)	14.89	5.02	4.97	10.11
O3 Soft Prompt (ours)	16.38	6.10	5.96	10.79
mBLIP‡	77.53	30.66	21.62	40.37

‡Scores reproduced from Geigle et al.; not directly comparable.

Table 6. Contextual comparison (CIDEr).

7. Conclusion

This work investigates whether linguistic typology can serve as a useful conditioning signal for multilingual image captioning. Typological structure is found to be learnable: both FiLM parameters and prompt representations reflect meaningful cross-lingual similarities. However, this does not directly translate into improved generation performance. Encoder-level language conditioning provides a strong baseline, prompt-based conditioning yields modest gains on seen languages, and FiLM, while structurally sound, introduces instability when unconstrained. Zero-shot transfer remains ineffective across all variants. These findings highlight three key limitations. First, the visual encoder remains English-aligned, creating a fundamental ceiling that lightweight conditioning cannot overcome. Second, training data is both limited and distributionally narrow, leading to template-driven generation that typology alone cannot correct. Third, the small number of training languages restricts coverage of the typological space, forcing extrapolation rather than interpolation for zero-shot languages. It is further observed that FiLM shift parameters (β) contribute negligibly, suggesting that a simpler γ-only formulation may suffice. Future work should address these limitations along three complementary directions. Increasing typological coverage through a more diverse set of training languages (not necessarily more data per language) may enable more reliable transfer. Aligning the visual encoder with multilingual generation, for example through Q-Former retraining as in mBLIP, is likely necessary to remove the current architectural bottleneck. At the same time, parameter-efficient strategies such as few-shot adaptation of language embeddings or prompt tokens provide a practical path to extending support to new languages. Finally, improving cultural and visual diversity in training data remains essential to move beyond template-driven captioning. More broadly, negative results here are informative. Three concrete failure modes are identified: unconstrained feature scaling, insufficient typological coverage, and encoder misalignment, each with a clear path forward. At the same time, the finding that FiLM recovers typological structure (correlation with URIEL) from only seven training languages without explicit supervision suggests that the underlying approach is valid. It is hoped that this work mo-

tivates more principled use of typological signals and better dataset construction for scalable low-resource multilingual captioning. The code is shared in a public repository.[1]

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[1] https://github.com/mammadov-aslan/typology-guided-multilingual-captioning

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